Vaccination to Prevent Disease Emergence

Can we stop vaccinating against pox viruses?

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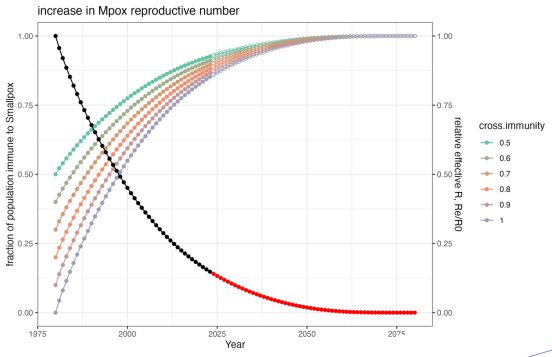
Motivation

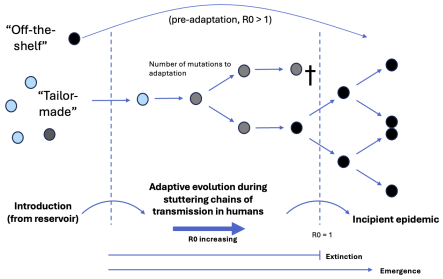
Vaccination to control highly transmissible infectious diseases is well-established and based on simple mathematic models of disease transmission in partially immune populations. The successful implementation of a vaccination strategy based on this principle of herd immunity led to the global eradication of Smallpox in 1980.



Because Smallpox has no known zoonotic reservoir, vaccination could end after eradication.

As herd immunity wanes, the reproductive numbers (REs) of related zoonotic orthopoxviruses, including Mpox, are likely increasing.

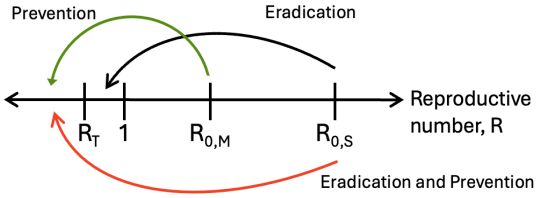
Though these viruses demonstrate limited human to human spread, there is concern that increasing **RE** could result in both **larger numbers of human cases and longer chains of transmission in humans, facilitating its adaptation to humans and its emergence as a highly transmissible pathogen** that has sustained spread in the human population



Approach and Takeaways

We developed a multi-type branching process model of disease transmission and evolution during long-chains of transmission in human populations. We used this model to explore **the efficacy of vaccination to prevent the adaptive evolution and emergence of a highly transmissible pathogen from spillover infections** with near-critical reproductive numbers.

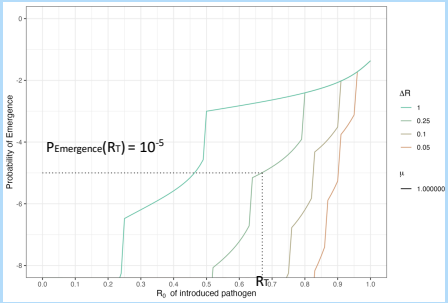
We define a framework analogous to **1-1/R0** to inform vaccination strategies to prevent disease emergence. This is given as $F_p = 1-R_T/R_0$.



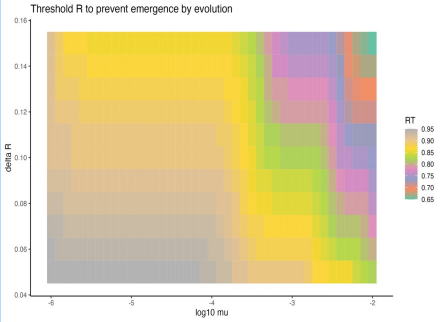
We find that models levels of vaccination can dramatically reduce the probability of disease emergence by adaptive evolution in transmission chains.

Results and Conclusions

Probability of evolving sustained transmission (R > 1) increases sharply with RE of the spillover infection (R_e < 1).



As R_e increases, fewer mutations are required to evolve R > 1, introducing sharp increases in emergence probability. Delta R: proportional increase in transmissibility with mutation; mu: rate (per infection) of adaptation.

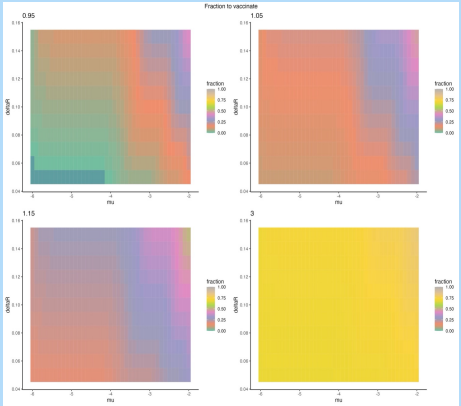


There exists some R_e < 1, R_T, above which evolution of sustained transmission during an outbreak may occur in the absence of intervention.

The fraction of population immunity required to prevent diseases emergence by adaptive evolution is given as:

$$F_p = 1-R_T/R_0.$$

Relatively little vaccination (< 30%) can dramatically reduce the probability of evolving sustained transmission for a large range of pathogen traits.



Preventing the emergence of zoonotic disease requires constant vaccination to maintain herd preventative immunity. This involves **significantly less vaccination** than eradication once endemic.